

The Problem

The confusions which occupy us arise when language is like an engine idling, not when it is doing work.

—LUDWIG WITTGENSTEIN¹

There is nothing so absurd that it cannot be believed as truth if repeated often enough.

—WILLIAM JAMES²

A dream we dream together is reality.

—BURNING MAN 2017

The mystery is always in the middle. I learned this wisdom early as a course instructor in the Medical School of the University at Pécs, Hungary. In my neurophysiology seminars, I enthusiastically explained how the brain interacts with the body and the surrounding world. Sensory stimuli are transduced to electricity in the peripheral sensors, which then transmit impulses to the midbrain and primary sensory cortices and subsequently induce sensation. Conversely, on the motor arc, the direct cortical

1. “Philosophy is a battle against the bewitchment of our intelligence by means of our language” (Wittgenstein: *Philosophical Investigations*, 1973). See also Quine et al. (2013). “NeuroNotes” throughout this book aim to remind us how creativity and mental disease are intertwined (Andreasen, 1987; Kéri, 2009; Power et al., 2015; Oszcoz-Iruezqui and Ortuño, 2016). [NeuroNote: Wittgenstein, a son of one of Austria’s wealthiest families, was severely depressed. Three of his four brothers committed suicide; Gottlieb, 2009].

2. <http://libertytree.ca/quotes/William.James.Quote.7EE1>.

pathway from the large pyramidal cells of the primary motor cortex and multiple indirect pathways converge on the anterior horn motor neurons of the spinal cord, whose firing induces muscular contraction. There was a long list of anatomical details and biophysical properties of neurons that the curriculum demanded the students to memorize and the instructors to explain them. I was good at entertaining my students with the details, preparing them to answer the exam questions, and engaging them in solving mini-problems. Yet a minority of them—I should say the clever ones—were rarely satisfied with my textbook stories. “Where in the brain does perception occur?” and “What initiates my finger movement, you know, before the large pyramidal cells get fired?”—were the typical questions. “In the prefrontal cortex” was my usual answer, before I skillfully changed the subject or used a few Latin terms that nobody really understood but that sounded scientific enough so that my authoritative-appearing explanations temporarily satisfied them. My inability to give mechanistic and logical answers to these legitimate questions has haunted me ever since—as it likely does every self-respecting neuroscientist.³ How do I explain something that I do not understand? Over the years, I realized that the problem is not unique to me. Many of my colleagues—whether they admit it or not—feel the same way. One reason is that the brain is complicated stuff and our science is still in its young phase, facing many unknowns. And most of the unknowns, the true mysteries of the brain, are hidden in the middle, far from the brain’s sensory analyzers and motor effectors. Historically, research on the brain has been working its way in from the outside, hoping that such systematic exploration will take us some day to the middle and on through the middle to the output. I often wondered whether this is the right or the only way to succeed, and I wrote this book to offer a complementary strategy.

In this introductory chapter, I attempt to explain where I see the stumbling blocks and briefly summarize my alternative views. As will be evident throughout the following chapters, I do believe in the framework I propose. Some of my colleagues will side with me; others won’t. This is, of course, expected whenever the frontiers of science are discussed and challenged, and I want to state this clearly up front. This book is not to explain the understood but is instead an invitation to think about the most fascinating problems humankind can address. An adventure into the unknown: us.

3. The term “neuroscientist” was introduced in 1969 when the Society for Neuroscience was founded in the United States.

ORIGIN OF TODAY'S FRAMEWORK IN NEUROSCIENCE

Scientific interest in the brain began with the epistemological problem of how the mind learns the “truth” and understands the veridical, objective world. Historically, investigations of the brain have moved from introspection to experimentation, and, along this journey, investigators have created numerous terms to express individual views. Philosophers and psychologists started this detective investigation by asking how our sensors—eyes, ears, and nose—sense the world “out there,” and how they convey its features to our minds. The crux of the problem lies right here. Early thinkers, such as Aristotle, unintentionally assumed a dual role for the mind: making up both the *explanandum* (the thing to-be-explained) and providing the *explanans* (the things that explain).⁴ They imagined things, gave them names, and now, millennia later, we search for neural mechanisms that might relate to their dreamed-up ideas.⁵

As new ideas about the mind were conceived, the list of things to be explained kept increasing, resulting in a progressive redivision of the brain's real estate. As a first attempt, or misstep, Franz Joseph Gall and his nineteenth-century followers claimed that our various mental faculties are localized in distinct brain areas and that these areas could be identified by the bumps and uneven geography of the skull—a practice that became known as *phrenology* (Figure 1.1). Gall suggested that the brain can be divided into separate “organs,” which we would call “regions” today. Nineteen of the arbitrarily divided regions were responsible for faculties shared with other animals, such as reproduction, memory of things, and time. The remaining eight regions were specific to humans, like the sense of metaphysics, poetry, satire, and religion.⁶ Today, phrenology is ridiculed as pseudoscience because we know that bumps on the skull have very little to do with the shape and regionalization of the brain. Gall represented to neuroscience what Jean-Baptiste Lamarck represented to

4. Aristotle (1908). We often commit similar mistakes of duality in neuroscience. To explain our results, we build a “realistic” computational model to convince ourselves and others that the model represents closely and reliably the “thing-to-be-explained.” At the same time, the model also serves to explain the biological problem.

5. A concise introduction to this topic is by Vanderwolf (2007). See also Bullock (1970).

6. Gall's attempt to find homes for alleged functions in the body was not the first one. In Buddhism, especially in Kundalini yoga, “psychological centers” have been distributed throughout the entire body, known as chakras or “wheels.” These levels are the genitals (energy), the navel (fire, insatiable power), heart (imaginary of art, dreams), the larynx (purification), mystic inwardly looking eye (authority), and the crown of the head (thoughts and feelings). The different levels are coordinated by the spine, representing a coiled serpent, in a harmonious rhythmic fashion. See also Jones et al. (2018).

preconceived ideas?⁷ But even with this recent expansion of brain regions, there are still many more human-concocted terms than there are cortical areas. As I discuss later in the book, cognitive functions usually arise from a relationship between regions rather than from local activity in isolated areas. But even if we accept that interregional interactions are more important than computation in individual areas, searching for correspondences between a dreamed-up term and brain activity cannot be the right strategy for understanding the brain.

THE PHILOSOPHICAL ROOTS

If cognitive psychology has a birthdate, it would be 1890, when *The Principles of Psychology*, written by the American philosopher and psychologist William James, was published.⁸ His treatment of the mind–world connection (the "stream of consciousness") had a huge impact on avant garde art and literature, and the influence of his *magnum opus* on cognitive science and today's neuroscience is hard to overestimate. While each of its chapters was an extraordinary contribution at the time, topics discussed therein are now familiar and acceptable to us. Just look at the table of contents of that 1890 work:

Volume 1

Chapter IV—Habit

Chapter VI —The mind–stuff theory

Chapter IX— The stream of thought

Chapter X—The consciousness of self

Chapter XI—Attention

Chapter XII—Conception

Chapter XIII—Discrimination and comparison

Chapter XIV—Association

Chapter XV—Perception of time

Chapter XVI—Memory

Volume 2

Chapter XVII—Sensation

Chapter XVIII—Imagination

7. Brodmann (1909); Glasser et al. (2016). For light reading about a recent quantitative craniocopy analysis, see Parker Jones et al. (2018).

8. James (1890).

Chapter XIX—The perception of “things”

Chapter XX—The perception of space

Chapter XXI—The perception of reality

Chapter XXII—Reasoning

Chapter XXIII—The production of movement

Chapter XXIV—Instinct

Chapter XXV—The emotions

Chapter XXVI—Will

Over the years, these terms and concepts assumed their own lives, began to appear real, and became the *de facto* terminology of cognitive psychology and, subsequently, cognitive neuroscience.

Our Inherited Neuroscience Vocabulary

When neuroscience entered the scene in the twentieth century, it unconditionally adopted James’s terms and formulated a program to find a home in the brain for each of them (e.g., in human imaging experiments) and to identify their underlying neuronal mechanisms (e.g., in neurophysiology). This strategy has continued to this day. The overwhelming majority of neuroscientists can pick one of the items from James’s table of contents and declare, “this is the problem I am interested in, and I am trying to figure out its brain mechanisms.” Notice, though, that this research program—finding correspondences between assumed mental constructs and the physical brain areas that are “responsible” for them—is not fundamentally different from phrenology. The difference is that instead of searching for correlations between mind-related terms and the skull landscape, today we deploy high-tech methods to collect information on the firing patterns of single neurons, neuronal connections, population interactions, gene expression, changes in functional MRI (fMRI), and other sophisticated variables. Yet the basic philosophy has remained the same: to explain how human-constructed ideas relate to brain activity.

Let’s pause a bit here. How on earth do we expect that the terms in our dreamed-up vocabulary, concocted by enthusiastic predecessors hundreds (or even thousands) of years before we knew anything about the brain, map onto brain mechanisms with similar boundaries? Neuroscience, especially its cognitive branch, has become a victim of this inherited framework, a captive of its ancient nomenclature. We continue to refer to made-up words and concepts and look for their places in the brain by lesion studies, imaging, and other methods. The alleged boundaries between predetermined concepts guide the search, rather than the relationships between interactive brain processes. We

identify hotspots of activity in the brain by some criteria, correlate them with James's and others' codified categories, and mark those places in the brain as their residences. To drive home my point with some sarcasm, I call this approach "neo-phrenology."⁹ In both phrenology and its contemporary version, we take ideas and look for brain mechanisms that can explain those ideas. Throughout this book, I refer to this practice as the "outside-in" strategy, as opposed to my alternative "inside-out" framework (Chapters 3, 5 and 13).

That brings me to the second needed correction in the direction of modern neuroscience: the thing-to-be-explained should be the activities of the brain, not the invented terms. After all, the brain gives rise to behavior and cognition. The brain should be treated as an independent variable because behavior and cognition depend on brain activity, not the other way around. Yet, in our current research, we take terms like emotions, memory, and planning as the to-be-explained independent categories. We will address these issues head on when we discuss the role of correlations and causation in neuroscience in Chapter 2.

The Outside-In Program

How has the outside-in program come to dominate modern neuroscientific thought? We can find clues in James's Table of Contents. Most of the titles have something to do with sensation and perception, which we can call the inputs to the brain coming from the outside. This is not by chance; William James and most early psychologists were strongly influenced by British empiricism, which in turn is based on Christian philosophies. Under the empiricist framework, knowledge derives from the (human) brain's ability to perceive and interpret the objective world. According to the empiricist philosopher David Hume,¹⁰ all our knowledge arises from perceptual associations and inductive reasoning to reveal cause-and-effect relationships.¹¹ Nothing can exist in the mind without first being detected by the senses because the mind is built out of sensations of the physical world. He listed three principles of association: resemblance, contiguity in time and place, and causation. For example, reading a poem can make you think of a related poem. Recalling one thing (cause) leads another thing to

9. Poldrack (2010) also sees a parallel between phrenology and the new fMRI mapping approach. However, his suggested program for a "search for selective associations" still remains within the outside-in framework.

10. [NeuroNote: Hume had several "nervous breakdown" episodes in his life.]

11. Associational theories assume that sequential events, such as our memories and action plans, are built by chaining associated pairs of events (*a* with *b*, *b* with *c*, etc). However, neuronal sequences (Chapter 7) contain important higher order links as well.

be remembered (effect). Western science and its methods are based largely on the inductive methodology that descended from Hume's perception-centered philosophy.

In the empiricism-inspired model, signals enter the brain from the outside, neuronal circuits process and perceive them, and some part of the brain decides whether or not to generate a motor response. The main emphasis is on perceptual processing and association, which are believed to be the main drivers of the brain's ability to comprehend and represent the outside world. In John Locke's words: "Ideas are clear, when they are such as the Objects themselves from whence they were taken."¹² Locke and his followers assumed that our sensors provide veridical descriptions of the objective and, I would add, observer-independent, world. An inevitable consequence of this perception-centered view is the assumption of an intelligent *homunculus* (i.e., a conscious selector, or, in modern-day neurospeak, "decision-maker").¹³ Indirectly, it also leads to the thorny problem of a free will that links perception to action. When "consciousness" is mentioned in experimental neuroscience circles, eyebrows are still raised today. "Decision-making," on the other hand, has become a ubiquitous, colloquial term. This is interesting since decision-making is defined as a process of collecting pertinent information, identifying alternatives, estimating the consequences, weighing the evidence, making choices, and taking actions.¹⁴ This definition is virtually identical to Thomas Aquinas's philosophical formulation of free will, except that his freedom of the self is granted to humans only, and they must make good choices (i.e., those that God likes).¹⁵ In contrast, making decisions is the one forbidden thing in Indian Buddhism.¹⁶

Because of the empiricist emphasis on sensory inputs and stimulus associations as the sole source of knowledge, it is not surprising that James dedicated only a few pages to action in his book. In this "outside-in" framework, the

12. Locke (1690); as quoted in Wootton (2015).

13. Several other related terms, such as "supervisor" and "executive" or "top-down functions," refer to the same fundamental idea.

14. Gold and Shadlen (2007); Kable and Glimcher (2009); Shadlen and Kiani (2013).

15. MacDonald (1998). More recently, decision-making has been popularized as a computer science metaphor, which regards past, present, and the future as separate entities. In this framework, memories belong to the past and decisions are made in the "now," with the expectation of benefits in the future (see Chapter 10).

16. In far Eastern philosophies, the concept of "freedom" (Chinese *tzu-yu*; Japanese *jiyu*) refers to liberation from the human nexus or being away from others. More recently, decision-making has been popularized as a computer science metaphor, one that regards past, present, and the future as separate entities. In this framework, memories belong to the past and decisions are made in the "now" with the expectation of benefits in the future (see Chapter 10).

brain is fundamentally a passive device whose main job is to perceive, evaluate the sensory inputs, weight their importance, and then—and only then—decide whether and how to act. As you will see, I disagree with this perception-action “outside-in” model (Chapters 3 and 5).¹⁷

Following the empiricists’ influence forward in time, we find that other important streams of thought in the nineteenth and twentieth centuries also adopted the outside-in approach. German Gestalt psychology, which has its own epistemological roots, also focused mainly on perceptual issues. *Cognitivism*, an offshoot of Gestalt psychology, was primarily interested in mental representation. It emphasized the input side of the brain and downplayed the importance of studying the output, although reaction time was often used to probe cognition.

The influence of the outside-in empiricist framework is also evident in Russian psychology, led by Ivan Petrovich Pavlov, even though such a connection has never been acknowledged in official Soviet scripts.¹⁸ Pavlov regarded the brain as an associative device. Learning in his dogs required only pairing of the conditioned and unconditioned signals, such as sound and food. As a result of repeated associations between them, the previously ineffective sound of the metronome comes to generate salivation (i.e., the conditioned reflex), which previously only the food was able to elicit (the unconditioned reflex). According to Pavlov, these associations are made automatically, which is why he called them *reflexes*. His dogs were constrained by a harness for practical reasons. Their task was to attend and associate, but no behavior was needed except to salivate and eat the food. Pavlov’s brain theory of classical conditioning is known as *stimulus substitution*; in essence, it assumes that, after conditioning the animal, the brain treats the conditioned signal in the same way as the unconditioned signal. Really? Is a dog not smart enough to understand that the metronome is not palatable?¹⁹

17. The current debate on consciousness also originates from the “brain as an input-processor” model (Dennett, 1991). According to Joaquín Fuster, “All goal directed behavior is performed within the broad context of the perception-action cycle” (Fuster, 2004). The explicit formulation of the perception-action cycle can be traced back to Jakob von Uexküll (“sensory-motor function circle”), who also coined the term *umwelt* (i.e., perceptual world). However, in contrast to the empiricist associationism, von Uexküll emphasized that the animal modifies its *Umwelt* (i.e., the surrounding world in which an organism exists and acts). von Uexküll introduced 75 new terms to explain his views (von Uexküll, 1934/2011). Situated, embodied cognition (Beer, 1990; Brooks, 1991; Noë, 2004), as an alternative for the associational models, grew out from this framework (Clark and Chalmers, 1998).

18. Pavlovian conditioning and the related dogma of imposed effects on the passively associating brain were a perfect fit for the ideology of the totalitarian Soviet system.

19. This short paragraph is an unfair treatment of the enormous value of the Pavlovian conditioning paradigm for the selfish aim of getting to my point quickly. Classical conditioning

Most contemporary computational models of brain function also fall under the outside-in framework (Chapter 13). Of these, perhaps the most influential is the model worked out by the British neuroscientist and thinker David Marr. He suggested that investigation of brain function should follow a three-level sequence of stages. First is a computational stage that defines the problem from an information processing perspective. The second, algorithmic stage should ask what representations the brain uses and how it manipulates those representations. Only after the first two stages are established can we move to the implementational stage; that is, how the algorithmic-mathematical model is solved at the level of synapses, neurons, and neural circuits.²⁰ I respectfully disagree with Marr's strategy. My first problem is the definition of "information processing" (see later discussion). Second, ample evidence is available to demonstrate that numerous algorithmic models are compatible with a given computational problem, but only one or a few are compatible with those implemented by neuronal circuits.²¹ Third, learning, the most fundamental aspect of the brain, is largely missing from his outside-in analysis.²² In contrast to the computational-algorithmic-implementation strategy, I argue throughout this book that understanding brain function should begin with brain mechanisms and explore how those mechanisms give rise to the performance we refer to as perception, action, emotion, and cognitive function.

Stimulus, Signal, and Reinforcement

A transient challenge to the outside-in framework was *behaviorism*. John B. Watson, the founder of behaviorism, attempted to position psychology as "a purely objective experimental branch of natural science" whose goal is the "prediction and control of behavior." His follower, B. F. Skinner, developed the

produced a plethora of interesting and important observations, including higher order conditioning, extinction, stimulus generalization and discrimination, associative strength, conditioned suppression, blocking, masking, latent learning, etc. The mathematical formulation of Rescorla and Wagner (1972) can account for many of the observations and yielded quantitative fits to a variety of experimental data. Yet stimulus substitution as an explanation for conditioning has persisted to date. Even Watson agreed: "The [conditioned stimulus] now becomes a substitute stimulus—it will call out the [response] whenever it stimulates the subject" (Watson, 1930; p. 21).

20. Marr (1982).

21. Prinz et al. (2004). In fact, it is hard to find an impactful neuroscience experiment which is based on Marr's suggested three-stage strategy.

22. As pointed out by Tomaso Poggio in his *Afterword* for the 2010 addition of Marr's *Vision* book.

theory of operant conditioning and suggested that behavior is solely determined by its consequences, depending on whether behavior leads to reward or punishment. Like Pavlov, the utopian views of Skinner advocated the *tabula rasa* or blank slate view of the brain, which can be shaped in any direction with appropriately dosed reinforcement. Although behaviorism departed from the assumption of passive associations and emphasized the importance of quantifiable motor behavior, it treated the brain as a “black box,” which may explain why cognitive neuroscience did not embrace its philosophy and practice until recently.²³

To explain why certain objects are more attractive than others, behaviorists introduced a hypothetical concept they called a *reinforcer* as an attempt to replace the homunculus decision-maker with something more tangible. Instead of the brain-resident homunculus or the unconditioned stimulus in Pavlovian conditioning, the selection is now made by the reinforcing positive and negative values of the reward and punishment, respectively. This is also an outside-in framework in the sense that explanation of behavior depends on outside factors.

Stimulus, signal, and reinforcer typically mean very different things to an experimenter. But do brains make such distinctions as well? Inputs from the environment and body arrive to the brain through the sensory interfaces. We often call these inputs “stimuli” or “signals.” Both terms imply a significant relationship between events outside the brain and the brain’s response to them. The term “stimulus” means that it exerts some change in brain activity because it must stimulate it. The term “signal” implies that the brain regards its change in response to the stimulus as important because it signals something useful to it. Thus, both the stimulus and signal refer to a relationship between the outside world and brain activity. Conversely, the absence of such relationship implies that the input does not exist from the point of view of the brain. Myriads of things in the surrounding world and our body remain unnoticed by our brains either because we do not have sensors for them (e.g., radio waves) or simply because they have no momentary significance (e.g., the T-shirt I am wearing while writing these sentences). There is nothing inherent in the physical things or events that would qualify them a stimulus or signal. Only when they are noticed by the brain do they become signals.

Now let’s turn to reinforcement. In the parlance of behaviorist literature, the reinforcer refers to a special quality in a stimulus that brings about a change in the animal’s behavior. Relatedly, reinforcement is the mechanism that increases or decreases the probability of recurrence of actions that lead to the positive

23. Watson (1930); Skinner (1938).

(pleasant) or negative (unpleasant) reinforcers, respectively. To register the occurrence of the reinforcer, the brain must perceive it. Thus, the reinforcing aspect of an input can be communicated only through sensory channels. Again, whether something is a reinforcer or not reflects a relationship between “things out there” and brain activity. The reinforcing quality is not inherent in any object or event. A chocolate may be a positive reinforcer, but if someone is forced to eat many pieces, it soon becomes aversive. If something is not noticed by the brain, it is irrelevant from the brain’s point of view; it does not bring about a change in neuronal activity.

The reader may have noticed that the content of the preceding two paragraphs is identical, even though they seemingly refer to different things: signal and reinforcer. Perhaps they mean the same thing. Some inputs induce large changes in behavior while most others bring about small or unnoticeable effects. The reinforcer or reward is just another regular stimulus that acts through the exact same sensory channels (smell, taste, or pain) as the stimuli, but it may exert a stronger impact on brain activity. Viewed in this way, signals and reinforcers represent a continuous distribution on a wide scale in which the two ends of the distribution are designated as positive or negative and those close to the middle as neutral signals. I devote two chapters to the discussion of broad distributions of many things in the brain and discuss that often the discrete nature of words simply reflects large quantitative differences (Chapters 12 and 13).²⁴

SHORTCOMINGS OF THE OUTSIDE-IN APPROACH

Ever since John von Neumann’s comparison between the computer and the brain, computer metaphors have dominated the language of neuroscience. A metaphor is a powerful tool to convey an idea because it relates a mysterious phenomenon to an understood one. However, metaphors can also be misleading because they may give a false sense of understanding a novel phenomenon before it is actually known how the thing works. In the brain–computer metaphor, neuronal operations are likened to the machines’ ability to transduce a

24. What determines the magnitude of the brain’s reaction to a stimulus? Over the past decades, the term “reinforcer” got separated from the signal and acquired a new meaning. In contemporary neuroscience, it is identified with *dopamine*, a neuromodulator produced in the ventral tegmental area and substantia nigra. An entire new field with strong links to computational neuroscience has emerged, known as “reinforcement learning,” with dopamine as the reinforcer or supervisor that assigns value to external stimuli. How stimuli can mobilize the dopaminergic neurons to alter and magnify the value of stimuli post hoc, the so-called “credit assignment problem,” is currently strongly debated. Reinforcement-learning algorithms are inspired by behaviorism (Sutton and Barto, 1998; Schultz, 1998, 2015).

real-world thing into symbols, called *representations*.²⁵ By this logic, symbols in the brain—the firing patterns of neurons, for example—should correspond to some real-world stimuli in the same way that computer algorithms correspond to, or represent, inputs from the objective world. However, unlike in the computer, the same stimuli often induce highly variable responses in the brain depending on brain states and testing conditions²⁶ (Chapter 6).

We often pretend that we know which features interest the brain, and correlate them with neuronal activity. Such arbitrariness is appropriate and, in fact, essential for symbolic representations by machines because they are trained to produce a “best match” between the experimenter-chosen features and symbols.²⁷ But brains are different. Representation of the objective world and finding truth are not their main jobs. The brain evolved not to represent anything but to help its host body to survive and reproduce. In addition, brains produce many effects that do not correspond to anything tangible in the physical world. Pleasure and fear, desire, irrational numbers or the entire discipline of math, and even Santa Claus feel real without corresponding to something veridical outside the brain.²⁸

According to the outside-in framework, the best strategy to understand mechanisms of perception and cognition is to present various stimuli (e.g., an object or aspects of an object) and examine the spatiotemporal distribution of the evoked neuronal responses in the brain (Figure 1.2).²⁹ To increase the rigor of the discovery process, initially “simple” stimuli are presented, such as vertical and horizontal gratings for vision or beeps of particular frequencies for hearing research. After the data are collected and interpreted, researchers can search for

25. *Representation* is a Platonic concept, referring to the idea that we humans cannot comprehend the real world (Kant’s *noumenon*) but only a representative of it (*phenomenon*). Both in classical Cartesian dualism and Lockean empiricism, the mind relies on the representation of *noumena* (Skarda and Freeman, 1987; Berrios, 2018).

26. A similar distinction was made between representation and meaning by Walter Freeman in his book *How Brains Make Up Their Minds* (1999). See also Eliasmith and Anderson (2003).

27. A digital camera can reliably “represent” what it “sees” in a different format (bits), but such information has no meaning to the camera. The many smart programs associated with picture compression, face recognition, and segmentation can reveal many important and useful statistical relationships about the inputs. In fact, by knowing the inputs and the computed results, one can effectively figure out the algorithms and even the electronic circuits involved. But this strategy often fails in the investigation of the brain (Chapter 3).

28. James (1890); Milner (1996); Damasio and Damasio (1994); LeDoux (2014, 2015).

29. Engel et al. (2001); Hebb (1949); James (1890); Milner (1996); von der Malsburg (1994); Hubel and Wiesel (1962, 1974); Rieke et al. (1997).

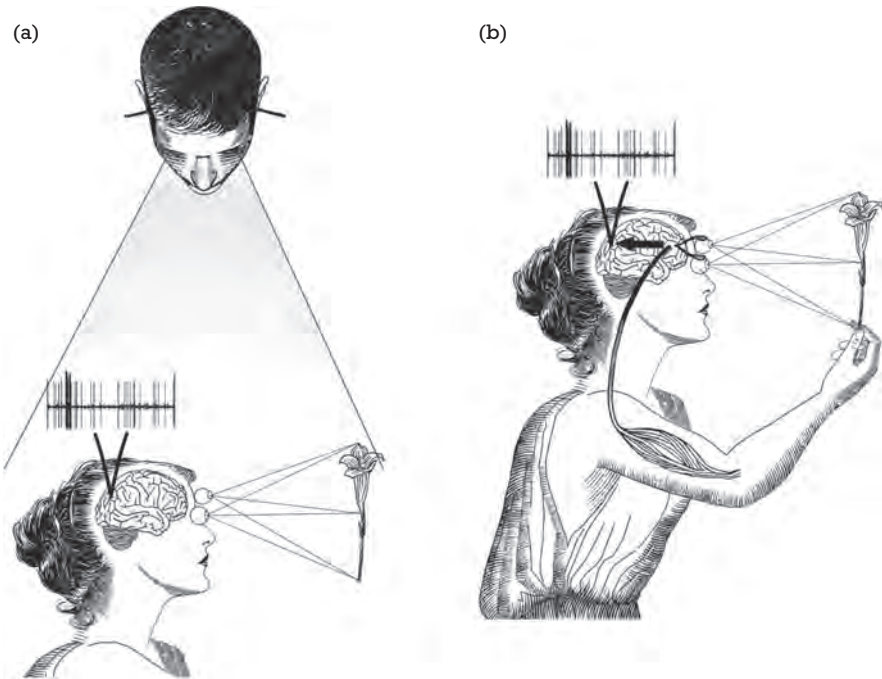


Figure 1.2. Outside-in and inside-out strategies. A: Outside-in. A stimulus is presented (flower) and responses (e.g., action potentials of a single neurons) are recorded from the sensory cortex. The experimenter has access to both the stimulus and the neuronal responses and can establish a reliable relationship between the input stimulus and neuronal activity. However, neurons in the sensory cortex do not “see” the stimulus. All they can register is action potential–induced inputs from other upstream neurons. B: Inside-out. The meaning of the stimulus can be grounded by comparing it something else. In addition to inputs from the retina, sensory neurons also receive input from movement-producing neurons in the motor area (called a “reafference” signal; discussed in detail in Chapter 3). Whenever a motor command is sent to the eye muscles and the arm, sensory neurons are informed about such outputs (*arrow*). By moving the eyes or moving the flower by hand, the visual neurons can get a “second opinion” about the flower. These action-induced signals are critical for providing meaning to the stimulus.

neuronal responses to increasingly complex stimuli. This strategy anticipates that (eventually) one should be able to explain how an object is represented in the brain from the observed relationship between the stimuli and the neuronal responses.

This research program was pioneered by the legendary scientific duo, David Hubel and Thorsten Wiesel, the Nobel Prize winners in Physiology and Medicine in 1981, who introduced single-neuron recordings to study the visual system. They made a long series of extraordinary discoveries that continue to influence

sensory research today. The theoretical framework of a passively associating brain was a welcome strategy to experimenters because neuronal responses to sensory stimuli could be registered also in anesthetized or paralyzed animals. When experiments were subsequently performed in waking animals, their heads and even the eyes were fixed to eliminate inconvenient behavior-induced variability in the recorded brain signals. However, building a complex percept in the brain from its assumed elementary components turned out to be a formidable hurdle given the astronomical realm of possible combinations.³⁰ Luckily, it turned out that natural visual scenes and sounds represent only a small subset of all possible combinations, so sensory researchers recently began to use natural images and sounds. They discovered that brains are “fine-tuned” for naturally occurring stimuli, and thus many aspects of neural activity were missed with the previous artificial stimuli. For example, even in a small brain, like that of a frog, transmission of acoustic information is most efficient when the spectrum of sounds matches the spectrum of natural frog calls.³¹ But the nature of the stimulus is only a technical problem. The bigger issue is that the brain is not a passive device that unconditionally associates signals.

A tacit assumption of the outside-in³² framework is that attributes or features of objects are bound together by some brain mechanism. What makes an object an entity is the coherent fusion of its components, which separates it from its surroundings. For example, a car has a particular shape with a body and wheels, color, sound, and a characteristic movement pattern. But there is a problem with this philosophy, which is that the “attributes” of an object are not a fixed set that reside in the object but are constructed in the brain of the observer. Brain responses do not intrinsically “represent” stimuli until the experimenter interprets the data.³³ Judgments of “similar” or “different” are subjective (i.e., observer-dependent). Yes, an experimenter can quantitatively define the relationship between stimuli and the spatiotemporal firing patterns of recorded neurons (Figure 1.2). However, this relationship does not mean that neurons in

30. An illuminating review on how methods influence scientific thinking is that by Evarts (1973). I often wondered how many more recordings, monkeys, and cats would be needed to explain vision with the outside-in strategy, even if all 30,000 neuroscientists were working on the visual system.

31. Attneave (1954); Singh and Theunissen (2003); Wang et al. (1995); Rieke et al. (1995).

32. Related terms to the “outside-in” framework are “empiricist,” “associational,” and “representational” strategies, and their various combinations. *Principles of Neural Science*, a 1,700-page “bible” of neuroscience and the most widely used textbook for educating neuroscientists, also follows the outside-in framework (Kandel et al., 2012).

33. Skarda and Freeman (1987); Werner (1988).

the brain of the experimental subject views the stimuli in the same way as the experimenter, nor does it mean that the brain uses these recorded signals in the same way or at all. This is the crux of the problem of the outside-in correlational approach. It cannot yield adequate understanding of perception or neuronal computation. I briefly summarize the outline of this argument here and return to it in more detail in Chapter 3.

ONLY ACTION CAN PROVIDE MEANING TO BRAIN ACTIVITY

Like the philosophical issues that plagued physics through the twentieth century, the issue in neuroscience is the “observer.” There is a fundamental difference between an observer outside the brain, such as the experimenter, and an observer inside the brain, such as Descartes’s hypothetical homunculus residing in the pineal gland.

First, let’s take the perspective of an experimenter studying the visual system. Upon presentation of a visual stimulus, an image is converted to action potentials by retinal ganglion cells and conveyed to the primary visual cortex via neurons in the thalamic lateral geniculate body. The experimenter can place a recording device, such as an electrode array, inside the brain and monitor neural responses to different images presented to the eye. In this situation, the experimenter has a privileged point of view: he has access to both the images (external world) and the neuronal responses in the brain (Figure 1.2). From this perspective, he can find a reliable correlation between specific features of the presented images (input; the rose in Figure 1.2) and neuronal responses (output for present purposes), which allows him to make a guess about the possible input–output transformation rules. After establishing a correlation (often mistakenly called a *neuronal code*), he can recover stimulus properties from the neuronal responses and even predict the properties of untested stimulus patterns (e.g., another flower). In short, a particular constellation of neuronal spike patterns *becomes* information. But to whom is that information useful? The recorded signal in the brain, combined with a priori knowledge about the stimulus, can generate meaning only to the experimenter. The observed correlation between the image and neuronal response only means that something outside the brain was transformed into firing pattern changes.

What about the internal observers, the brain circuits? The neurons that “respond” to the image do not see the image, just patterns of neural spiking arriving from their peer neurons. Neurons in the visual cortex are blind to events that happen in the world and so is the rest of the brain or even a magical homunculus (i.e., the little human) watching the action of all visual cortical neurons

(Figure 1.2). Imagine that the little human is the experimenter who, in a magic experiment, can register the activity of all neurons in the brain. Without access to information outside the skull, such knowledge cannot reveal what the brain is sensing; it cannot see the rose. In the language of the neural code, the outside-in framework has shown only that decoding of stimulus properties from neural activity is possible *in principle* and only *if an observer has the code book* (i.e., the set of stimulus–response correlations). However, the brain only has its own neural “responses,” and the outside-in framework does not mention how such a code would be generated or read by neural responses alone.

All any neuron in the brain ever “sees” is that some change occurred in the firing patterns of its upstream peers. It cannot sense whether such change is caused by an external perturbation or by the brain’s perpetual self-organized activity (Chapters 7 and 8). Thus, neurons embedded in networks of other neurons do not “know” what the brain’s sensors are sensing³⁴; they simply respond to their upstream inputs. In other words, unlike our ideal experimenter, the neurons have no way of relating or *comparing* their spikes to anything else because they only receive retinal correspondences or processed “representations” of the sensory input. But establishing correspondences without knowing the rules by which those correspondences are constructed is like comparing Mansi words with Khanty words when we understand neither language.³⁵ Only after we have defined the vocabulary of one language can we understand the corresponding meaning of words in the other. Similarly, without further information, sensory neurons can attach no meaning whatsoever to their spikes. Put simply, the mind’s eye is blind.³⁶

To explain my point differently, let me invite you to do a thought experiment. Let’s connect the output of a video camera (or the optic nerve of the retina) to cultured neurons in a dish, derived from the visual cortex of the smartest person in the world, so that a fraction of the neurons is stimulated by the

34. A classic paper on this topic is by Lettvin et al. (1959). See also Eggermont (2007).

35. This is not to say that everything is equal to a newborn brain. Phylogenetic experience can provide the necessary cipher for certain patterns, such as human faces, for babies. But, for most of the things in the world, we have to create a cipher through action-based grounding.

36. This “blindness,” in my view, applies also to the Global Workspace version (Baars, 1988; Dehaene and Changeux, 2011) of the Cartesian theater metaphor of consciousness, since the Global Workspace, even if it involves distributed processing in the entire brain, faces the exact same problem: having no access to the external information. Realizing the shortcomings of the Cartesian theater idea, Dennett and Kinsbourne (1992) suggested an alternative solution, what they call the “Multiple Drafts” model, reminiscent of E. Roy John’s distributed consciousness model (1976). However, this model is also a “passive” interpreter and has no access to outside world information and, therefore, faces the same “grounding problem” (Harnad, 1990).

camera output. The part of the tissue culture that is stimulated by the camera output can be called the *sensory region*. We may discover that repeated visual patterns, such as the picture of the rose, will induce somewhat similar neuronal responses. We may even detect some changes in synaptic connectivity among neurons as a result of repetition. It is also possible, though, that some members of the cultured neuronal network will be preoccupied with each other and ignore the stimulation altogether.³⁷ The experimenter can establish a relationship between the applied stimuli and the corresponding neuronal responses and declare that some coding was identified. But the tissue culture does not “see” the rose. So far, this is analogous to the passive, representational brain model and the experimental approach used in many laboratories. Since the neurons in the culture have no direct access to the outside world, they have no way of verifying or “grounding” their patterns to the events outside. The term “grounding” refers to the ability of the brain’s circuits to connect it to something meaningful for the owner of the brain. It’s kind of a second opinion from a more reliable source.³⁸ Using our dictionary metaphor, if you know the meaning of words in an English dictionary, you can ground the meaning of corresponding words in a dictionary of any other language.

The situation in the dish may change dramatically if the spiking output of some neurons is connected to a robot that can move the camera based on the spiking pattern of these experimenter-designated “motor” neurons in the culture. Now neurons in the dish have two functions. One function is generating an output (move the camera) and the second is responding to the camera-supplied signals (sense the inputs). Because the tissue culture circuit is supposedly well interconnected, neurons in the “sensory” region will receive two types of inputs: input from the camera signals and an additional input from their motor neuron partners, which inform them that it was their action that induced movement of the camera and, therefore, changes in the sensory input. Now, we have a closed-loop

37. It helps if the tissue culture is “awakened” by adding a cocktail of subcortical neurotransmitters, mimicking the aroused state of the brain (Hinard et al., 2012).

38. Steve Harnad (1990) explains the need of “symbol grounding” in understanding. An effective computer algorithm can discover correspondences between two sets of symbols without understanding the meaning of any of them. Handheld translators have enormous vocabularies and can convert commonly spoken words and phrases to dozens of other languages at an amazing speed and even pronounce the words in the target language—all done by a few chips and a microprocessor. However, the semantic “meaning” of symbols of one set (e.g., a new language) can be constructed only if the translator’s knowledge is grounded to an already known set of symbols (e.g., your mother tongue). Without grounding, a theorem remains disconnected from empirical phenomena. See also Fields (2014). Grounding is also related to the “credit assignment problem”; that is, the determination of how collective action of neurons relates to behavior.

system in a dish, an output-driven sensory feedback device. The addition of the robot provided the cultured neurons with the same advantage as the experimenter: now the neurons can sense both the outside world and the internal computation and, therefore, can compare their joint impact.³⁹

Now let's assign a goal to our closed-loop system—for example, to find the rose in a picture. Whenever the camera is focused on the rose by the chance behavior of the cultured neurons, a magic potion (let's call it dopamine) is sprayed onto the culture to strengthen the connections between those neurons that successfully moved the camera to the desired spot in the visual scene and the neurons that were activated by the rose. This simple modification (“plasticity”) may increase the probability that such neuronal activity will happen again and lead to pointing the camera at the rose more often.⁴⁰ Thus, out of the large realm of possible neuronal network patterns, at least one unique event acquired a meaning. By connecting a bunch of otherwise useless tissue culture neurons with the world through an action-perception arc, we have just built a machine with a purpose: a brain-like device.⁴¹ In fact, we may not need a magic potion if we assume that the brain is already prewired to some extent so that its connectivity is not random but guided by some evolution-shaped statistical rule (Chapter 11). The internal feedback from the action-inducing neurons to sensory neurons may be regarded as a second opinion, the needed grounding signal that can attach meaning to input-induced neuronal responses (Chapter 3).

39. For an attempt to achieve such comparison by neurons, see Demarse et al. (2001). A computational model using such building blocks is by Weng (2004); Choe et al. (2007).

40. Artificial intelligence (AI) devices using cameras give equal priority to all image pixels at the earliest stage of processing. In contrast, the visual system prioritizes and isolates the most relevant information based on previous knowledge and moves the eyes to attend those aspects (Olshausen et al., 1993). Inspired by the brain's solution, recent AI architectures serially take glimpses of the input image and update their internal state representations before sampling the next location. The goal of AI agents is to select the best outcome that results in maximal “reward” (Mnih et al., 2014). The main success of AI agents, in, for example, defeating human players in Atari games, is that those games are also designed by humans using artificial algorithms. The current instantiations of AI agents are still pretty helpless when it comes to developing and solving temporally extended planning strategies.

41. William Grey Walter at the Burden Neurological Institute in Bristol, England built an electronic “tortoise,” among the first autonomous robots, which he called *Machina speculatrix*. His *Machina* had only two “neurons” (vacuum tubes) but could be assigned a goal, which was to find its way to a recharging station when it ran low on battery power. Walter (1950) argued that goal-directed action is a main characteristic of animals and that animated gadgets with a goal, such as his robots, can emulate such behaviors. A similar argument was made subsequently by Brooks (1991): brains have no representations. Instead, “it is a collection of competing behaviors. Out of the local chaos of their interactions there emerges, in the eyes of an observer, a coherent pattern of behavior.”

In our tissue culture example, we arbitrarily designated two sets of neurons and called them *sensory input* to where the stimuli from the camera were delivered and *motor output* because these neurons were connected to the robot moving the camera. The tissue culture has no a priori bias toward these designations, and the two areas are reciprocally interconnected to each other. In real brains, the sensory and motor areas have somewhat different internal connectivity but the similarities are more striking than the differences. The differences are even more blurred when we compare their partner structures, often referred to as *higher order sensory* and *supplementary motor areas*.⁴² These areas are reciprocally connected, and the traffic moves not only from the sensory to the motor circuits but equally so from the motor to the sensory areas. This latter motor-to-sensory projection has a fundamental impact on many levels of brain operations, as we will see in the subsequent chapters.

The picture I just painted is unfortunately an oversimplification. I did not mean to create the impression that our tissue culture thought experiment can explain the mechanisms of a thinking, feeling brain. It was only meant to illustrate the minimum necessary requirements of a brain-like system. Random connections and plasticity are not enough (Chapter 12). A large part of brain structure is genetically determined, and self-organized activity is as important for the development of brain dynamics as is sensory stimulation even in primary sensory areas of the brain⁴³ (Chapter 5).

THE INSIDE-OUT, READER-CENTRIC FRAMEWORK

Can James's Table of Contents and the outside-in strategy be renegotiated? I do not think it would get us far. My point is not that measuring brain responses

42. Mrganka Sur and colleagues (Sharma et al., 2000) "rewired" the brain of ferrets so that the visual information was connected to the auditory system. Neurons in the auditory cortex of these animals responded in many ways like the visual cortex of intact animals, including visual orientation modules. Impressively, the ferrets used their newly wired brain to avoid visual objects.

43. Brain circuits can assemble themselves under genetic guidance to provide "proto-maps." When the thalamic input to the neocortex is disrupted in Gbx-2 mutant mice, neocortical region-specific gene expression and the main cortical layers and divisions develop normally (Miyashita-Lin et al., 1999). Deletion of a single protein, Munc18-1, in mice abolishes neurotransmitter secretion from synaptic vesicles throughout development. However, this does not prevent an apparently normal initial assembly of the brain. However, persistent absence of synaptic activity leads to synapse degeneration and the mice die (Verhage et al., 2000). The constellation of ion channels in individual neurons is also driven by their activity.

to stimuli yields no valuable data or insight. In fact, a primary slogan in my laboratory is that “it is impossible to find nothing” in the brain. An enormous amount of precious knowledge has been obtained by identifying neuronal “correspondences” and “representations” of the sensory world over the past decades. Until recently, the empiricist outside-in approach has generated most of the available knowledge about sensory coding, and it continues to be an important strategy. However, when studying the more central parts of the brain involved in cognition, the limitations of a strict stimulus–response strategy become apparent.⁴⁴ The ultimate goal of brain research cannot just be to uncover correspondences between external signals and neuronal responses. Building a dictionary of representations from scratch would be an enormous effort, but it is easy to see why such a program has limited success (Chapter 4). The outside-in approach interrogates brain circuits by asking what those circuits can *potentially* do, rather than what they actually do, which produces an illusory understanding of brain computation. Overall, I suggest that searching for alternative approaches is warranted.

The essence of the complementary strategy I suggest is simple: understand the brain from within. In this inside-out framework, the key issue is the mechanisms by which stimuli and situations become meaningful percepts and experiences for the brain.⁴⁵ The main emphasis is on how the brain’s outputs, reflected by the animal’s actions, influence incoming signals. By linking an otherwise meaningless brain pattern to action, that pattern can gain meaning and significance to the organism. Within the brain, the emphasis is on how downstream networks make use of the messages obtained from their upstream partners. In this framework, the goal of the brain is to explore the world and register the consequences of successful exploratory actions to improve the efficacy of future actions. Thus, an action–perception loop learns to make sense of sensory inputs. Perception is what we do (Chapter 3).⁴⁶

44. Rieke et al. (1997); Friston (2010, 2012).

45. The outside-in versus inside-out dichotomy should not be confused with the top-down and bottom-up distinctions. The former refers to the relationship between the world and the brain, whereas the latter refers mainly to the spread of activity in anatomical space. It is also distinct from Paul MacLean’s (1970) concept of the *triune brain*, a division into reptilian, paleomammalian, and neo-mammalian brain. The relationship of the inside-out framework to embodied cognition and predictive coding will be addressed in Chapters 3 and 5.

46. In my younger days, I thought these ideas were solely mine. But, with time, I discovered that many people, often independently from each other, have come to a similar conclusion (e.g., Merleau-Ponty, 1945/2005; Mackay, 1967; Bach-y-Rita, 1983; Held and Hein, 1983; Paillard, 1991; Varela et al., 1991; Berthoz, 1997; Bialek et al., 1999; Järvillehto, 1999; O’Regan and Noë, 2001; Llinás, 2002; Noë, 2004, 2009; Choe et al., 2007; Chemero, 2009; Scharnowski et al.,

Nearly all chapters of this book revolve around an action-centered brain because I want to make it as clear as possible that no meaning or advantage emerges for the brain without the ability to calibrate neural patterns by behavior-induced consequences. Without generating outputs to move and optimize the sensors, and calibrate brain circuits at some point in our lives, I claim, no perception or cognition exists.⁴⁷

A major advantage of inside-out approach is that it is free of philosophical connotations. It takes brain mechanisms as independent variables, as opposed to attempting to find correspondences between subjectively derived categories and brain responses. Brains largely organize themselves, rather than being enslaved to input signals. I am interested in how internally generated, self-organized patterns in the brain acquire “meaning” through action, which becomes what we call “experience.” Instead of looking for correlations between experimenter-selected signals and activity patterns of neurons, we should ask a more important question: How do those firing patterns affect downstream “reader” neurons? We can call this strategy the *neural observer*, or *reader-centric*, view. Before you accuse me of simply replacing the homunculus with the “reader,” please consult Chapters 4 and 5. The term “reader/observer” is meant to be an engineering term, referring to an actuator (such as a robot, muscle, or simply a downstream neuron) that responds to a particular pattern of upstream neuronal activity but not to others; only a particular key of a bunch on a keychain opens a particular lock. In the subsequent chapters, I will give you a taste of how this is done in the brain by discussing a set of recent findings that illustrate progress toward an inside-out research program.

2013). Interaction between the body and mind was natural to the Greek philosophers. Aristotle might have said that we learn about the world by our actions on it. Only after the concept of the soul fused with the mind came the full separation, especially with Descartes’s declaration of the independence of *res extensa* (corporeal substance or the external stuff) and the God-given *res cogitans* (the thinking thing or soul). Perhaps Henri Poincaré was the first thinker who speculated that the only way sensations become relevant and turn into an experience is by relating them to the body and its movements. “To localize an object simply means to represent oneself the movements that would be necessary to reach it” (Poincaré, 1905; p. 47). The volume by Prinz et al. (2013) is an excellent update on the renewed interest in action science. Tolman’s cognitive map theory (1948) is another important and major departure from the outside-in framework but it is not action-based.

47. Once a brain pattern acquires meaning (i.e., it has been calibrated through action), it remains meaningful as long as memory is intact, without further action (Chapters 5 and 8). Learning through action also applies to species knowledge acquired through evolutionary selection (Watson and Szathmáry, 2016).

BLANK PAPER OR PREEXISTING CONSTRAINTS?

The choice of outside-in and inside-out, action-centric frameworks also shapes our ideas about fundamental brain operations. One of the oldest outside-in views was formulated by Aristotle, who suggested that we are born with a blank slate (or *tabula rasa*) on which experiences are written.⁴⁸ The *tabula rasa* view is an almost inevitable assumption of the outside-in framework because, under the empiricist view, the goal of the brain is to learn and represent the truth with its veridical details. This view has influenced thinking in Christian and Persian philosophies, British empiricism, and the Marxist doctrine, and it has become the leading school of thought in cognitive and social sciences.⁴⁹ Similar to the free will idea that humans can do anything, *tabula rasa* thinking implies that anything can be written on a blank brain. Although I have never met a neuroscientist colleague who would openly subscribe to the *tabula rasa* view, many experiments and modeling studies today are still performed according to this associationist philosophy-based framework, or its modern-day version: “connectionism.” The main idea of connectionism, especially in cognitive psychology and artificial intelligence research,⁵⁰ is that perception and cognition can be transcribed into networks of simple and uniform neurons, which are interconnected relatively randomly with largely similar synaptic connections. Nothing is further from the facts, as I explain in Chapter 12 and briefly summarize here.

Early behavioral observations already argued against the idea of the brain as a blank slate. Researchers demonstrated repeatedly that animals do not associate everything equally and cannot be trained to do all tricks the experimenter expects them to do. Behaviors that relate to the animal’s ecological niche can be trained easily because the brain is predisposed or “prepared” to do things that have survival and reproductive advantage. For example, “spontaneous alternation,” the tendency in rodents to choose different paths during foraging, is an instance of biological preparedness for the rapid acquisition

48. The *tabula rasa* concept was more explicitly developed by the Persian Muslim Avicenna (Abu Ali al-Hussain Ibn Sina). Avicenna was an important precursor of British empiricists. His syllogistic, deductive method of reasoning influenced Francis Bacon, John Locke, and John Stuart Mill and continues to influence scientific reasoning even today. Steven Pinker’s *Blank Slate* (2003) is an excellent overview on this old philosophical problem.

49. Read, for example, Popper (1959).

50. The classic works on connectionist networks have been summarized in Rumelhart et al. (1986) and McClelland et al. (1986). The much more sophisticated recent neuronal networks also fall under the *tabula rasa* framework. The leading author on connectionism in cognitive psychology is Donald Hebb (1949).

of species-specific learning. Returning to the same location for food within a limited time window is not an efficient strategy because choosing an alternate route will more likely lead to reward. In contrast, associations that would be detrimental to survival are called “contraprepared.” For example, it is virtually impossible to train a rat to rear on its hindlimbs to avoid an unpleasant electric shock to the feet since rearing is an exploratory action and incompatible with the hiding and freezing behaviors deployed in case of danger.⁵¹ A practical message here is that if training takes weeks or longer, the experimenter should seriously consider that the brain signals associated with such extended shaping of behavior may not reveal much about the intended question because each animal may use different tricks to solve the task, and these may remain hidden for the experimenter.

Neuroscience has also accumulated much experimental evidence against the *tabula rasa* model. The most important is the recognition that a great deal of activity in the brain is self-organized instead of being driven by outside signals. This self-generated persistent activity is supported by the numerous local and brain-wide neuronal rhythms, which we will discuss in Chapter 6. For now, it is sufficient to note that these oscillations are not only part of and help to stabilize neuronal dynamics but also offer a substrate for syntactical organization of neuronal messages. Each period of an oscillation can be conceived as a frame that contains a particular constellation of spiking neurons. Metaphorically, we can call it a “neuronal letter.” In turn, the cycles of the many simultaneously acting oscillations can concatenate neuronal letters to compose neural words and sentences in a virtually infinite number of ways. This brain syntax is the main focus of the discussion in Chapters 6 and 7.

51. Each of these telegraphic statements represents large chapters of cognitive and behavioral psychology. Skinner believed that any behavior could be shaped by his method of “successive approximations” and was able to teach rodents and other animals to do some extraordinary things. However, his students (Breland and Breland, 1961) countered that animals often misbehave and develop “autoshaped” or “superstitious rituals” (e.g., Catania and Cutts, 1963). The literature on autoshaping is a rich source of information for the expected surprises that today’s users of the increasingly popular “virtual reality” tasks will have to face (Buzsáki, 1982). The prepared and contraprepared spectrum of behaviors, developed by Martin Seligman, had a great influence on human phobia research (Seligman, 1971, 1975). Phobias (e.g., to spiders, snakes, and heights) can be viewed as a prominent prepared category because they are commonly associated with objects and situations that have threatened us throughout our evolutionary history. Joe LeDoux, the leading authority on threat behavior, amplifies this view further by explaining that brain circuits that detect and respond to threats in rodents are not necessarily the same as those that produce conscious fear in humans (LeDoux, 2015).

Wide-Range Distributions: Good Enough and Precise Solutions

As an alternative to the empiricist outside-in *tabula rasa* view, I raise the possibility that the brain already starts out as a nonsensical dictionary. It comes with evolutionarily preserved, preconfigured internal syntactical rules that can generate a huge repertoire of neuronal patterns. These patterns are regarded as initially nonsense neuronal words which can acquire meaning through experience. Under this hypothesis, learning does not create brand new brain activity patterns from scratch but is instead a “fitting process” of the experience onto a preexisting neuronal pattern.

The preconfigured brain is supported by a mechanism I refer to as “high-diversity” organization (Chapters 12 and 13). Diversity and large variations are the norm in the brain, whether we are analyzing its microscopic or macroscopic connectivity or dealing with its dynamics at a small or a large scale. Neuronal firing rates, connection strength between neurons, and the magnitude of their concerted action can vary by three to four orders of magnitude. I speculate that this diversity, which respects a logarithmic rule (Chapter 12), is the essential backbone that provides stability, resilience, and robustness to brain networks against competing requirements for wide dynamic range, plasticity, and redundancy. There are many advantages of a preconfigured brain⁵² over the blank slate model, the most important of which is stability of brain dynamics because adding new experience after each learning event does not perturb much the overall state of neuronal networks. I will discuss that even the inexperienced brain has a huge reservoir of unique neuronal patterns (Chapter 13), each of which has the potential to acquire significance through the experience of unique events or situations.

The tails of these wide-range distributions, such as neurons with high and slow firing rates and strong and weak synaptic connections, can support seemingly opposing functions, such as judging a situation as familiar or novel. Neural circuits involving a minority of highly interactive and well-connected neurons may appear as a constraint. However, the benefit of such apparent constraint is that the collective action of such oligarchic circuits with skewed distribution of resources allows the brain to generalize across situations and offer its “best guess” under any circumstances, so that no situation is regarded

52. “Preconfigured” usually means experience-independent. The backbone of brain connectivity and its emerging dynamics are genetically defined (Chapter 12). In a broader sense, the term “preconfigured” or “preexisting” is also often used to refer to a brain with an existing knowledge base, such as mine at this instant of time after numerous years of action-based calibration.

as completely unknown. I call this aspect of neuronal organization the “good-enough” brain. The good-enough brain uses only a small fraction of its resources, but its member neurons are always on the alert. On the other hand, the remaining majority of slow-firing member neurons with weak synaptic connections have complementary properties, such as high plasticity and slow but precise and reliable processing. This other extreme of network organization comprises the “precision” brain, whose main job is to redefine a situation or an event and determine its distinctness in high detail. The “good enough” and “precision” aspects of brain organization complement, and in fact, assume, each other (Chapters 12 and 13).

In the preconfigured brain model, learning is a *matching* process, in which preexisting neuronal patterns, initially nonsensical to the organism, acquire meaning with the help of experience. As discussed earlier, attaching meaning or significance to any neuronal pattern requires grounding, which can be provided only by action, the ultimate source of knowledge (Chapter 3). In contrast to the *tabula rasa* model, in which the complexity of neuronal dynamics scales with the amount of experience, in the preconfigured brain network homeostasis is only minimally and transiently affected by new learning because nothing is added to the circuitry; only rearrangements take place. The total magnitude of plasticity in brain circuits and the overall spiking activity remain constant over time, at least in the healthy adult brain (Chapter 12). I make no claim that my strategy is definitely right, but the alternative formulation that I am suggesting might provide different insights than the current representational framework.

In contrast to the preconfigured brain, the outside-in framework assumes that there are obvious things in the outside world, which we often call information, and the brain’s job is to incorporate or absorb that information. But how?

THE BRAIN “CODES INFORMATION”: OR DOES IT?

This often-heard statement should communicate a lot of content to everyone. After all, we live in an Information Age in which everyone is familiar with such terms as “information” and “coding.” But when we attempt to explain and define information coding in the brain, we pause because we do not have a widely accepted, disciplined definition. When terms are not defined precisely, they often acquire many disguised interpretations and, therefore, convey different meanings to different individuals. “Is my grant proposal rejected or approved?” From the perspective of information theory, the

answer represents a single bit of information (yes or no).⁵³ But for the receiver of such information (me), it induces profound and long-lasting changes in my brain.

While the mathematical formalism of information theory is neutral to information “content,” the same cannot be said for the brain. For a behaving organism, “decoding” information cannot be divorced from “meaning”: the experience-based prediction of potential consequences of that information. There is no such thing as invariant information content in a stimulus.⁵⁴ “Features” or attributes of a stimulus are not objective physical characteristics of the stimulus. Instead, the relevance of the stimulus depends on changing internal states and the brain’s history with similar situations. In the following chapters, I hope to show you that this situational relevance is not a small matter but has a profound and observable effect on the nature of neural activity and its relationship to the sensory environment. The same visual stimulus (e.g., the rejection letter) evokes very different neuronal patterns in a frog’s brain, my brain, or the brains of other humans. The induced pattern even in my brain depends on its current state. If the letter arrived the same week when I received the news about other rejections, it would induce a different effect compared to an alternative situation when the letter arrived a day after I won the lottery jackpot.

53. Once agreed by the sender and receiver agents, everything can be represented by symbols; for example, the head or tail outcome of coin flipping by 1 and 0. Each flip produces one bit of information, which can be concatenated by some agreed rules. In the American Standard Code for Information Interchange (ASCII), seven bits represents a character; for example, 1100101 is code for the letter “e.” Each block representing a single letter is separated by an agreed SPACE code (0100000). As in Morse code or any other coding scheme, spaces are the essential part of packaging messages (Chapter 6). Without space codes, the information may become completely undecodable. Physics uses information as a conceptual tool to understand interactions and relationships. Information can be mathematically defined as the negative of the logarithm of the probability distribution ($I = \log_2 N$). When $n = 2$, information I is 1, thus the unit of information is a choice between two possibilities, measured in “bits” (Shannon, 1948). The *meaning* of the observed events does not matter in this definition of information.

54. I accept that one solution would be to stick to the framework of Shannonian theory or entropy (Shannon, 1948) and use it to define relationships between world events and brain events. A connection to information theory is *free energy* that quantifies the amount of prediction error or surprise. *Prediction error* is the difference between the representations encoded at one level and the top-down predictions generated by the brain’s internal model: a comparison. More formally, it is a variational approximation to negative log likelihood of some data given an internal model of those data (Friston, 2010). However, in this definition, we should abandon the wider use of information as it is typically applied in neuroscience.

Computation performed by computer programs or machines is referred to, in general, as “information processing.”⁵⁵ In reality, the information is not in the processing. It *becomes* information only when interpreted by an observer that recognizes its meaning and significance, be it a human interpreter or a mechanical actuator. A computer program can effectively control a robot or other man-made machines, thus giving rise to the illusion that information resides in the program. However, the process is based on a human-designed solution even if it involves a complex trial-and-error learning process, including “deep learning” in artificial intelligence programs.⁵⁶ In short, information is not inherent in the computation (in machines or brains) but becomes such when it is interpreted.⁵⁷

In neuroscience, information is often understood as acquired knowledge, a broader concept than in information theory. Claude Shannon warned us that information theory is “aimed in a very specific direction, a direction that is not necessarily relevant to such fields as psychology.”⁵⁸ In the forthcoming chapters, I discuss why the brain is not an information-absorbing, perpetual coding device, as it is often portrayed, but a venture-seeking explorer, an action-obsessed agent constantly controlling the body’s actuators and sensors to test its hypotheses. The brain ceaselessly interacts with, rather than just detects, the external world in order to remain itself. It is through such exploration that correlations and interactions acquire meaning and become information (Chapter 5). Brains do not process information: they create it.

Now let me try to give a broad definition of coding as it applies to the brain. In essence, coding is an agreement for communication between the sender and receiver, making the content of the encoded information a secret to outsiders.

55. The term “processing” has an interpretative connotation. It refers to a procedure that takes something and turns it into something else, a sort of format transformation. This is what is usually referred to as “representation.” The often used “information-processing” expression relies on this view and tacitly implies that representation is the final product. However, representation is not a thing; it is a process or an event (Brette, 2015, 2017). Therefore, representation (and perception) cannot be described as a transformation of one thing into another thing. See also Jeannerod (1994).

56. Deep learning (or hierarchical deep machine learning or deep structured learning) is part of a broader family of machine-learning algorithms that attempts to model high-level abstractions in data by using a “deep graph” with multiple processing layers, akin to the associational networks sandwiched between sensory and motor areas in the brain. For the origin of this highly successful and rapidly expanding field, see Hinton et al. (1995) and Sejnowski (2018).

57. The thoughtful paper by Chiel and Beer (1997) outlines a broad program for brain-inspired robot design. See also König and Luksch (1998).

58. Shannon (1956).

If the code is not known to the receiver, the packets carry no inherent information (Chapter 7). There are many forms of coding (e.g., Morse code or the genetic code), but the fundamental features remain the same. The encoding–decoding process is done via a cipher,⁵⁹ a transforming tool that encrypts messages by mutually agreed syntactical rules, so the information looks like nonsense to the uninitiated. Encryption is not always intentional, of course; sometimes it just happens. Evolution has no aim of secrecy, yet biological codes are often mysterious to us—that is, until we uncover the cipher. In the outside-in philosophy, the words in James’s list are the senders and the brain is the receiver. However, terms such as “emotion,” “will,” “attention,” and others are not senders of anything and possess no inherent information. Space and time are not sensed directly by the brain either (Chapter 10), and no messages are being sent by them. If we want to understand information, we should start with its creator: the brain.

CODE-BREAKING EXERCISES

The American writer, Edgar Allen Poe, whose hobby was cryptography, believed that “human ingenuity cannot concoct a cipher which human ingenuity cannot resolve.” But modern science claims that it is possible to encrypt messages that no human or machine can break.⁶⁰ Does the cipher of brain dynamics fall into this highly secured category, or can it be outsmarted?

As we search for ciphers in the brain, we may learn from previous deciphering successes. The Englishman Thomas Young and the Frenchman Jean-François Champollion are credited with deciphering Egyptian hieroglyphics by using a written cipher: the Rosetta Stone. This most famous artifact of the British Museum contains the same script in three languages (Greek, Demotic, and hieroglyphics). As a polymath, Young’s strategy was to find correspondences of well-known names of gods and royalties, such as Ptolemy and Cleopatra, between the words in the Greek text and the glyphs. This approach provided fast initial progress but could not uncover the generative rules of hieroglyphic language. That task needed Champollion’s genius and his understanding of syntactic

59. From the Arabic *sifr*, meaning “zip” or “blank.” Deciphering is the process of breaking the secret (i.e., decoding). After reading extensively about ciphers, I realized that brain research badly needs such concepts. Yet very little specific research has tried to uncover and understand such mechanisms, even though we pay extensive lip service to “breaking the brain’s code.”

60. Poe (1841). An accessible review on the current state of ciphers is by Eckert and Renner (2014). The key is to make choices that are independent of everything preexisting and are hence unpredictable: a device-independent cryptography.

rules in Hebrew, Syriac, Greek, Arabic, Aramaic, Persian, and, critically, Coptic and Demotic. He hypothesized that phonetic signs were not just used for the names of royals but also represented a general rule. This insight suggested that Egyptian hieroglyphs could be read and understood by transforming the glyphs to sounds and concatenating them.⁶¹ Thus, the nonsensical messages of a forgotten language became information. However, without the Rosetta stone, the secrets of the Egyptian hieroglyphs might well have remained forever hidden.

My second example of a cipher is connected to the most celebrated (posthumously) code breaker, Alan Turing, a genius by many accounts. He and his team at Bletchley Park, north of London, are credited with breaking the code of the secret messages of the German Navy in World War II.⁶² With all my admiration and respect for Turing's contributions, I must make a correction: neither he nor his team "broke the code." They received the cipher, the key syntactical tool of encryption, in the form of Enigma machines,⁶³ from the Polish Cipher Bureau early during the war.⁶⁴ Without this access to the syntactical rules of the cipher, the secret messages sent to the German U-boats by the Kriegsmarine Headquarters would likely have remained gobbledygook to the allies. Alan Turing's Rosetta Stone was the Enigma's hardware.

Is deciphering a secret code possible without (some) a priori knowledge of the syntax? Our third example, discovery of the genetic code and the translation of DNA to RNA to protein, was such a case. The cipher in this case was the double spiral. The two strands of the helix consist of polynucleotides made up from simpler units of nucleotides. Each nucleotide is composed of a nitrogen-containing nucleobase—cytosine (C), guanine (G), adenine (A),

61. Silvestre deSacy, Champollion's master, had already found a relationship between pronounced sounds of Demotic symbols and their meaning (Allen, 1960). Champollion is often criticized for not giving enough credit to deSacy and Young. Eventually, 24 glyphs, representing single consonants, were identified, similar to an alphabet.

62. Turing's short and complex life continues to inspire generations of creative people. The movie *The Imitation Game* (2014) is perhaps the best known, albeit heavily romanticized, account of his code-breaking years and personal life.

63. If the letters are shuffled properly, as can be achieved today with fast computers, the cipher is practically unbreakable.

64. In a book about the history of British Intelligence in World War II, mathematician Marian Rejewski noted, "we quickly found the [wirings] within the [new rotors], but [their] introduction . . . raised the number of possible sequences of [rotors] from 6 to 60 . . . and hence also raised tenfold the work of finding the keys. Thus, the change was not qualitative but quantitative" (as quoted in the appendix of Kozaczuk, 1984). Another important predecessor of Turing's decoding venture was the work of another British cryptographer, Dillwyn Knox, who worked on deciphering Enigma-encrypted messages of the Nationalists in the Spanish Civil War (Batey, 2009).

or thymine (T)—as well as a monosaccharide sugar called deoxyribose and a phosphate group. The syntax is elegantly simple. A is always paired with T, and C paired with G, providing the rule for making the double-helix DNA: the hydrogen bonds between these matched pairs of nucleotides bind the nitrogenous bases of the two separate polynucleotide strands. This DNA sequence is ultimately translated into a string of amino acids, the building blocks that make up proteins. This is done via RNA, which transcribes a particular set of three nucleotides into a particular amino acid.⁶⁵ Using this messenger RNA as a template, the protein molecule is then built by linking neighboring amino acids.

These examples of code-breaking practices and their players offer an important lesson for brain scientists in our search for a neuronal “code.” When the syntax is known, breaking a code is a hopeful exercise. Uncovering the meanings of the hieroglyphs took a few thinkers and a few decades of hard work. A few years of nerve-wracking search for correspondences, with several hundred people working together, were required to decipher the messages encrypted by the Enigma machines, even though the cipher was available. In the case of the genetic code, with well-understood goals (a combination of four elements to code in triplets for 20 known amino acids), the efforts of several hundreds of scholars over the span of a few decades provided a mechanistic explanation for a “miracle” of nature, the hypothetical force that produces complex organisms from simple seeds. So what should be our strategy for identifying the brain’s cipher? Collecting correspondences and correlations is not enough; we also need to understand the neuronal syntax (Chapters 6 to 8) and to “ground” our correlational observations with a firmer “second opinion” (Chapters 3 and 5). Only with such independent knowledge can we hope to figure out the secret meanings of the brain’s vocabulary.

At this point, the reader may rightly ask: Why does it matter to logically proceed from inside out rather than the other way around? Can we not understand relationships either way? Answering this will require discussion of the complex problem of causality in neuroscience, the topic we are going to explore in the next chapter.

65. James Watson, Francis Crick, and Maurice Wilkins were acknowledged for their discovery of the DNA code with the Nobel Prize in 1962. The Russian physicist Georgiy Gamow had already calculated, using mathematical modeling, that a three-letter nucleotide code could define all 20 amino acids. He formed the “RNA Tie Club” with 20 regular members (one for each amino acid), and four honorary members (one for each nucleotide; Gamow, Watson, Crick, and Sydney Brenner) with the goal of wide cooperation in understanding how RNAs can build proteins. Yet the breakthrough came from outsiders, Marshall W. Nirenberg, Har Gobind Khorana, and Robert W. Holley, who were awarded the Nobel Prize “for their interpretation of the genetic code and its function in protein synthesis” in 1968. See <http://www.nobelprize.org/educational/medicine/gene-code/history.html>.

SUMMARY

In this introductory chapter, I presented some contrasting views and research strategies that can provide complementary knowledge about brain operations. The first comparison was between “outside-in” and “inside-out” strategies. The outside-in or perception-action framework has deep philosophical roots, starting with Aristotle and most explicitly elaborated by the empiricist philosopher David Hume. According to this framework, all our knowledge arises from perceptual associations and inductive reasoning of cause-and-effect relationships. This philosophy is the foundation of Western science and continues to influence thinking in contemporary neuroscience. An inevitable consequence of this perceptual representation-centered view is the assumption of a hidden homunculus that decides whether to respond or not. In contrast, my recommended inside-out framework takes action as the primary source of knowledge. Action validates the meaning and significance of sensory signals by providing a second opinion. Without such “grounding,” information cannot emerge.

Next, we discussed the relationship between the *tabula rasa* (blank slate) and preconfigured brain models. In the empiricist outside-in model, the brain starts out as blank paper onto which new information is cumulatively written. Modification of brain circuits scales with the amount of newly learned knowledge by juxtaposition and superposition. A contrasting view is that the brain is a dictionary with preexisting internal dynamics and syntactical rules but filled with initially nonsense neuronal words. A large reservoir of unique neuronal patterns has the potential to acquire significance for the animal through exploratory action and represents a distinct event or situation. In this alternative model, the diversity of brain components, such as firing rates, synaptic connection strengths, and the magnitude of collective behavior of neurons, leads to wide distributions. The two tails of this distribution offer complementary advantages: the “good-enough” brain can generalize and act fast; the “precision” brain is slow but careful and offers needed details in many situations.